Kerbo-S Test Plan, Sprint 2

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This plan defines 59 test cases that cover every Must requirement in Requirements Document v2.1, each with steps, an expected result and a status. It updates the Sprint 1 draft (57 cases, all Not Yet Tested) with the Sprint 2 results: **24 cases Pass, 0 Fail, 0 Blocked and 35 Not Yet Tested**, all of which are planned for Sprint 3. Two cases were added in Sprint 2 (TC-KC-15 and TC-UI-06). Ben recorded the Sprint 2 results and evidence (S2-23, PR #18). Every Pass is backed by evidence in the repository: the automated test log `docs/evidence/pytest-sprint2.txt` (68 tests passed, run Sat Nov 7 on the final main branch) and the screenshots in `docs/evidence/`. Two defects were found on real lab data and fixed before the final run (Section 6).

1. What Will Be Tested

In scope: data collection and parsing, the eleven audit rules (KC-01 to KC-11), risk scoring, database storage, AI Gateway safety and fallback behavior, reports and PDF export, core GUI flows, security controls, and the detection and AI quality metrics.

Out of scope: automated remediation, Windows hosts, Active Directory, real-time monitoring, load testing beyond lab size, and the general quality of third-party AI models beyond the Section 7.2 evaluation in the requirements.

2. Test Approach

Level	What it covers	Tool	When
Unit	One rule, parser, scorer or sanitizer at a time	pytest with fixtures	Every commit
Integration	Collect or import, parse, audit, score and store using snapshots	pytest and the CLI	Before each merge to the main branch

Level	What it covers	Tool	When
System (lab)	Live collection from the MIT Kerberos lab with planted misconfigurations (TV-01 to TV-15)	CLI and manual checks	End of each sprint
AI evaluation	Explanation and recommendation accuracy, consistency, usefulness	Evaluation script and reviewers	After the AI module exists
UI	Core screens and flows	Manual checklist	Before each demo

Ground truth. Each scenario has an `expected_findings` manifest written and committed before the audit runs. A script compares results to the manifests and computes detection rate, false-positive rate, false-negative rate and classification accuracy. In Sprint 2 the comparison is done by the end-to-end tests (TC-KC-12, TC-KC-15); the metric script is Sprint 3 (TC-EVAL-01).

3. Test Environment

Component	Detail	Status
KDC server	MIT Kerberos on Ubuntu Server 26.04 LTS, Proxmox VM `kerbo-kdc` (2 CPU, 4 GB RAM, 32 GB disk), realm KERBO.LAB, host kdc.kerbo.lab; setup in `docs/lab-setup.md`	Established Nov 5 (TC-ENV-01)
Kerberos client VM	Generates authentication activity	Not set up; in Sprint 2 `kinit` ran on the KDC itself. Sprint 3
Admin workstation	Python, PyQt6 and pytest on team machines: a Linux development VM (final test run, Python 3.14.4, pytest 9.2.0) and a Windows laptop (window screenshots)	Used for Sprint 2; dedicated Ubuntu desktop VM in Sprint 3

Component	Detail	Status
Test principals and fixtures	`tests/fixtures/real/clean`, `planted` and `aged`, captured from the KDC with `tools/capture_fixtures.sh`; each folder has an `expected_findings.yaml` written before the audit	Created Nov 6 to 7 for TV-01 to 04, 06, 07, 09, 10, 12; log scenarios Sprint 3
Mock AI provider	Returns fixed text for offline tests	Selected in `config/app.yaml`; not built (Sprint 3)
OpenAI test key	Used only for the AI evaluation	To obtain in Sprint 3

4. Entry Criteria, Exit Criteria and Status Values

- **Entry:** the code under test is merged to the main branch, and the fixture for the scenario exists.
- **Exit for Sprint 2:** every Must case for collection, rules and scoring that was in Sprint 2 scope is Pass, and any Fail has a logged defect with an owner. **Met:** every Sprint 2 case passed and both defects found were fixed.
- **Exit for the project:** all cases are Pass or have an accepted, documented deviation, and the Section 7 metric targets are met.
- **Status values:** Pass, Fail, Blocked (waiting on something else), Not Yet Tested.

5. Test Cases

5.1 Environment and fixtures

ID	Requirement	Steps	Expected result	Status	Actual Result	Date Run	Evidence
TC-ENV-01	A-01	Start the KDC VM, create a test user, run `kinit` and `klist` from the client	Ticket granted; `klist` shows a ticket-granting ticket	Pass	kinit alice and klist run on the KDC itself (Sprint 3 will include a separate client VM). klist shows a TGT for alice@KERBO.LAB, krbtgt/KERBO.LAB@KERBO.LAB	Nov 5, 2026	docs/evidence/TC-ENV-01.png

ID	Requirement	Steps	Expected result	Status	Actual Result	Date Run	Evidence
TC-ENV-02	TV-01 to TV-15	Create the planted principals, policies and config changes for each scenario; capture `kadmin` and log output	Fixtures saved with an `expected_findings` manifest per scenario	Pass	Fixtures for TV-03, 06, 07, 09, 10 captured from the KDC; aged copies for TV-01, 02, 04; manifests written before planting. TV-05 and TV-08 need KDC logs, planned for Sprint 3.	Nov 6, 2026	tests/fixtures/real/planted, tests/fixtures/real/aged
TC-ENV-03	FR-COL-08	Export a snapshot from the clean baseline realm (TV-12)	Snapshot folder with JSON, raw files and hashes	Not Yet Tested	Planned for Sprint 3		

5.2 Collection and parsing

ID	Requirement	Steps	Expected result	Status	Actual Result	Evidence
TC-COL-01	FR-COL-01	Parse principal fixtures for 10 principals	10 Principal objects with all fields populated and kinds classified	Pass	14 planted principals parsed with all fields and kinds (alice user, ben/admin admin, HTTP/web service, krbtgt internal); 3 tests passed	pytest-sprint2.txt
TC-COL-02	FR-COL-02	Parse `getpols` and `getpol` fixtures	Policy values equal the fixture values	Pass	Policies parsed: standard min length 12, weakpol 11; 2 tests passed	pytest-sprint2.txt
TC-COL-03	FR-COL-03	Parse `krb5.conf` and `kdc.conf` fixtures containing weak enctypes	Settings extracted with file, section, key and value	Pass	krb5.conf and kdc.conf settings extracted with file, section, key and value, including the planted rc4-hmac and allow_weak_crypto; 7 tests passed	pytest-sprint2.txt

ID	Requirement	Steps	Expected result	Status	Actual Result	Evidence
TC-COL-04	FR-COL-04	Parse `kadm5.acl` and the justification file	Admin entries with the correct justified flag	Not Yet Tested	Planned for Sprint 3	
TC-COL-05	FR-COL-05	Parse a KDC log fixture with successful and failed requests	Events with timestamp, principal, client, type and result	Not Yet Tested	Planned for Sprint 3	
TC-COL-06	FR-COL-06	Inspect the command allow-list and run collection with a command logger	Only list and get commands run; no modifying command is possible	Not Yet Tested	Planned for Sprint 3	
TC-COL-07	FR-COL-07, NFR-SEC-04, NFR-REL-01	Feed an unreadable file, a truncated log and a malformed principal line	Error recorded per source; remaining stages run; no crash	Pass	Malformed principal line, missing kdc.conf and bad config line each recorded as an error; rest loaded; no crash; 3 tests passed. Truncated log case planned for Sprint 3 (KDC logs)	pytest-sprint2.txt
TC-COL-08	FR-COL-08, FR-COL-09	Export a snapshot, then import it	Import succeeds, hashes verify, findings equal the live run	Not Yet Tested	Planned for Sprint 3	
TC-COL-09	TV-14	Change one raw artifact in a snapshot, then import	Hash mismatch reported and the audit flagged	Not Yet Tested	Planned for Sprint 3	

5.3 Audit rules

ID	Requirement	Steps	Expected result	Status	Actual Result	Evidence
TC-KC-01	KC-01, TV-01	Users with passwords 91 days and 90 days old	Finding for 91 days only; observed 91, threshold 90	Pass	91-day user flagged, 90-day not, service ignored (3 tests); aged: alice (91) flagged, dave (90) not	pytest-sprint2.txt
TC-KC-02	KC-02, TV-02, FR-AUD-05	Service principals at 181 and 180 days; a user at 181 days	Finding only for the 181-day service principal	Pass	181-day service flagged, 180 not, 181-day user not (3 tests); aged: HTTP/web (181) flagged, ldap/dir (180) not	pytest-sprint2.txt
TC-KC-03	KC-03, TV-03	Policy minimum length 11 and 12; one principal with no policy	Findings for length 11 and for the no-policy principal; none for 12	Pass	Length 11 flagged, 12 not; no-policy user flagged; built-ins skipped (4 tests); planted: weakpol and carol flagged	pytest-sprint2.txt
TC-KC-04	KC-04, TV-04	Users inactive 181 days and 179 days; one never authenticated, created 200 days ago	Findings for 181 days and for the never-authenticated user	Pass	181 days flagged, 179 and 180 not; never-authenticated user measured from password date (4 tests); aged: frank (181) flagged, dave (179) not	pytest-sprint2.txt
TC-KC-05	KC-05, TV-05	Service principal with no ticket request for 181 days; one with a recent request	Finding for the unused one only	Not Yet Tested	Planned for Sprint 3 (needs the KDC log parser)	
TC-KC-06	KC-06, TV-09	One admin principal missing from the justification file; one listed	Finding for the unlisted admin only	Pass	Unlisted admin flagged, listed not (2 tests); planted: eve/admin flagged, ben/admin not	pytest-sprint2.txt
TC-KC-07	KC-07, TV-06	Enable RC4 in `kdc.conf`; set `allow_weak_crypto = true`; add a principal with a 3DES key; also test an AES-only config	Findings for each weak setting and key; none for AES-only	Pass	RC4 in kdc.conf, allow_weak_crypto and an RC4 key flagged, AES-only not, DEPRECATED label handled (2 tests); planted: all 3 flagged. Used an RC4 key instead of the plan's 3DES key	pytest-sprint2.txt
TC-KC-08	KC-08, TV-07	Ticket lifetime of 25 hours and of 24 hours	Finding for 25 hours only	Pass	25 hours flagged, 24 not (1 test); planted: max_life 25h flagged, ticket_lifetime 24h not	pytest-sprint2.txt

ID	Requirement	Steps	Expected result	Status	Actual Result	Evidence
TC-KC-09	KC-09, TV-10	Renewable lifetime of 8 days and of 7 days	Finding for 8 days only	Pass	8 days flagged, 7 not (1 test); planted: max_renewable_life 8d flagged, renew_lifetime 7d not	pytest-sprint2.txt
TC-KC-10	KC-10, TV-08	10 failures in 15 minutes; 9 failures; 10 failures spread over 30 minutes	Finding only for 10 failures within 15 minutes	Not Yet Tested	Planned for Sprint 3	
TC-KC-11	KC-11, TV-11	Request burst above the threshold and one below	Finding only for the burst	Not Yet Tested	Planned for Sprint 3	
TC-KC-12	TV-12, FR-AUD-02, NFR-AUD-01	Run the clean baseline snapshot twice	Zero findings both times; identical output	Pass	Clean fixtures: zero findings on both runs, identical output	pytest-sprint2.txt
TC-KC-13	FR-AUD-07	Run an audit with no AI configured and the network disabled	Audit and scoring complete	Pass	Network disabled (unshare --net), AI provider mock: planted audit completed with 8 findings, all scored	pytest-sprint2.txt, TC-KC-13.png
TC-KC-14	TV-15	Run the compound scenario (one service principal with RC4, an old password and an unjustified admin entry)	KC-02, KC-06 and KC-07 all reported for that principal	Not Yet Tested	Planned for Sprint 3	
TC-KC-15	TC-ENV-02 manifests	Run the audit on the planted and aged fixtures and compare to expected_findings.yaml	Every expected finding present; nothing from not_expected; no extra findings	Pass	Planted: all 8 expected, no boundary flagged, no extras; aged: all 11; both saved to SQLite with every finding stored (4 tests)	pytest-sprint2.txt

5.4 Risk scoring and storage

ID	Requirement	Steps	Expected result	Status	Actual Result	Evidence
TC-RSK-01	FR-RSK-01, FR-RSK-02	Score each rule using its baseline factors	Scores match Requirements Section 6.2 (for example KC-07 is 88.5 Critical, KC-02 is 70.0 High)	Pass	All 8 Sprint 2 rules match Section 6.2: KC-01 38.5, KC-04 35.0, KC-08 42.0, KC-09 40.0 Medium; KC-02 70.0, KC-03 65.0, KC-06 68.0 High; KC-07 88.5 Critical (6 tests). KC-05, 10, 11 in Sprint 3	pytest-sprint2.txt
TC-RSK-02	FR-RSK-02	Score values 24.9, 25, 49.9, 50, 74.9 and 75	Low, Medium, Medium, High, High, Critical	Pass	Exactly as expected for all six values (6 tests)	pytest-sprint2.txt
TC-RSK-03	FR-RSK-03	Change one baseline in config and rerun	New score applied; factor values stored with the finding	Not Yet Tested	Planned for Sprint 3. Factor values are already stored with each finding (TC-DB-02)	
TC-DB-01	FR-DB-02	Try UPDATE and DELETE on a stored finding; update a review note	Both rejected for the finding; the review note update succeeds	Pass	UPDATE and DELETE on a finding rejected with "findings are immutable"; review note updated (2 tests, plus manual check)	pytest-sprint2.txt, TC-DB-01.png

ID	Requirement	Steps	Expected result	Status	Actual Result	Evidence
TC-DB-02	FR-AUD-04, FR-DB-01	Run an audit and read a finding back	Record has rule, object, observed value, threshold, timestamp, artifact reference and hash	Pass	Saved planted audit reads back with rule, object, observed, threshold, audit time and an artifact with a 64-character SHA-256; CLI saves each audit (2 tests)	pytest-sprint2.txt

5.5 AI Gateway

All cases in this section are planned for Sprint 3.

ID	Requirement	Steps	Expected result	Status
TC-AI-01	FR-AI-01, FR-AI-04, FR-AI-06	Run analysis with the mock provider	Fixed text stored and displayed, labelled AI-generated and advisory	Not Yet Tested
TC-AI-02	FR-AI-02, FR-AI-03	Capture the outgoing payload for a finding on a principal	Only allow-listed fields; principal replaced by a token; no raw log, key or description text	Not Yet Tested
TC-AI-03	TV-13, TM-02	Put prompt-injection text in a principal description and a log message, then run analysis	Text removed before sending; output unaffected; audit completes	Not Yet Tested
TC-AI-04	FR-AI-09	Provider returns invalid JSON, then valid; separately invalid twice	One retry then success stored; two failures give status unavailable	Not Yet Tested
TC-AI-05	FR-RSK-04	Provider reply includes a field that tries to set a score or severity	Field rejected; stored score unchanged	Not Yet Tested

ID	Requirement	Steps	Expected result	Status
TC-AI-06	FR-AI-05	Disable the network or key, run analysis and export a report	User notified; audit, scores and report complete without AI sections	Not Yet Tested
TC-AI-07	FR-AI-09	Run one finding and inspect the stored output	Model name, prompt version and timestamp stored	Not Yet Tested
TC-AI-08	FR-AI-08, TV-15	Analyze the compound scenario	One related group naming the three findings with a rationale; remediation order shown beside score order	Not Yet Tested

5.6 Reports

All cases in this section are planned for Sprint 3.

ID	Requirement	Steps	Expected result	Status
TC-RPT-01	FR-RPT-01, FR-AI-07	Generate the technical report for the TV-15 audit	Findings, scores, evidence references and AI text present	Not Yet Tested
TC-RPT-02	FR-RPT-02, FR-AI-07	Generate the executive report	Posture summary, counts by severity, top risks and compliance summary	Not Yet Tested
TC-RPT-03	FR-RPT-03	Export a PDF to a chosen folder; cancel; export to an unwritable folder	File opens; cancel leaves nothing; clear error for the unwritable folder	Not Yet Tested
TC-RPT-04	FR-RPT-04, FR-RPT-05	Generate a report, edit a finding row in a copy of the database, run verify	Digest mismatch reported for the copy; the original verifies	Not Yet Tested

5.7 User interface

ID	Requirement	Steps	Expected result	Status	Actual Result	Evidence
TC-UI-01	FR-UI-01, NFR-PERF-02	Start an audit from the GUI in snapshot mode	Progress shown; window stays responsive; results appear	Not Yet Tested	Planned for Sprint 3 (worker thread and progress bar)	
TC-UI-02	FR-UI-02	Open the dashboard with mixed findings; filter by severity and category	Sorted by score; filters work; counts match the database	Pass	Severity and category filters and counts work (5 tests); window screenshots taken by Jessica on Nov 6	pytest-sprint2.txt, TC-UI-02-clean-folder.png, TC-UI-02-sample-table.png
TC-UI-03	FR-UI-03, FR-UI-04	Open a finding, mark it reviewed and add a note	Detail shows all fields; status and note saved	Not Yet Tested	Planned for Sprint 3	
TC-UI-04	UC-01 alternate flow	Start a live audit with the KDC unreachable	Error shown; audit ends cleanly	Not Yet Tested	Moved to Sprint 3: needs live collection	
TC-UI-05	FR-UI-05	Run the CLI audit on a snapshot	JSON results written; exit code 0	Pass	CLI printed JSON with 2 findings, saved the audit, exit code 0 (sample snapshot)	TC-UI-05.png
TC-UI-06	FR-COL-07	Load a missing, an invalid-JSON and an incomplete snapshot from the CLI and the window	Clear message, non-zero exit code in the CLI, message box in the window; no crash	Pass	Missing file, invalid JSON, missing fields and bad principal each give a clear message and exit code 2, no traceback (5 tests)	pytest-sprint2.txt, TC-UI-06-invalid-json.png

5.8 Security and non-functional

All cases in this section are planned for Sprint 3.

ID	Requirement	Steps	Expected result	Status
TC-SEC-01	NFR-SEC-02	Create the database, a snapshot and a report; check permissions	Owner-only access (600 for files, 700 for folders)	Not Yet Tested
TC-SEC-02	NFR-SEC-03	Scan the repository and config for API keys; run with the key unset	No secrets found; app reports the missing key and continues without AI	Not Yet Tested
TC-SEC-03	NFR-SEC-01	Compare the Kerberos database before and after an audit	No change	Not Yet Tested
TC-PERF-01	NFR-PERF-01	Audit a snapshot with 500 principals and 100,000 log lines	Finishes in under 5 minutes excluding AI	Not Yet Tested
TC-USE-01	NFR-USE-01	A new user exports a PDF from a snapshot	Done in 5 actions or fewer	Not Yet Tested
TC-PORT-01	NFR-PORT-01	Install from `requirements.txt` on a clean Ubuntu VM and launch	Installs without manual fixes; application starts	Not Yet Tested
TC-MNT-01	NFR-MNT-01	Run the suite and check each rule has a passing and a failing case	Coverage check passes	Not Yet Tested

5.9 Metric evaluation

All cases in this section are planned for Sprint 3.

ID	Requirement	Steps	Expected result	Status
TC-EVAL-01	Requirements Section 8.1	Run all 15 scenarios and compute metrics against the manifests	Detection rate 95% or higher, false negatives under 5%, false positives under 10%, classification accuracy 95% or higher	Not Yet Tested
TC-EVAL-02	Requirements Section 7.2	Score 55 AI outputs (11 rules x 5 runs) plus TV-15	Accuracy 90% or higher, consistency 95% or higher, usefulness 4.0 or higher	Not Yet Tested

6. Defects and Known Issues

Two defects were found in Sprint 2. Both appeared only when the code met real output from the lab KDC, which is the reason the fixtures were captured from the lab rather than written by hand. Both were fixed and covered by tests before the final run, so no Sprint 2 test case failed and none is marked Fail.

Defect	Test case	Description	Severity	Status
DF-01	TC-KC-07, TC-KC-15	On the lab KDC, `getprinc` lists an RC4 key as `DEPRECATED:arcfour-hmac`. KC-07 compared the whole label, so the planted RC4 key on `host/legacy.kerbo.lab` was missed (a false negative). Reported by Ben, Fri Nov 6.	High (a planted weakness was not reported)	Fixed by Jessica in PR #13, Nov 6: KC-07 now checks each colon-separated part of the label. Test `test_tc_kc_07_deprecated_label_on_rc4_key_flagged` added.
DF-02	TC-COL-03, TC-KC-12	The lab `krb5.conf` contains an `includedir` line. The configuration parser treated it as a malformed line and recorded a parse error, so the clean realm did not load cleanly. Reported by Ben, Fri Nov 6.	Medium (a false error, no wrong finding)	Fixed by Jessica in PR #13, Nov 6: the parser skips `include`, `includedir` and `module` lines. Test `test_tc_col_03_include_directives_are_not_errors` added; the clean-realm test (TC-KC-12) now loads with no errors.

Known issues that may affect testing:

1. **Log formats unverified.** The KDC log path and message formats for KC-05, KC-10 and KC-11 are not yet confirmed in the lab (Sprint 3).

2. **Last-authentication data.** `getprinc` updates the last successful authentication only on initial (AS) requests, so KC-05 must use the KDC log. For KC-04, a never-authenticated user is measured from the last password change date, because `getprinc` prints no creation date.
3. **Aged dates are edited.** The aged fixture is a copy of the planted capture with six date lines moved back, because a new lab cannot produce 90-day-old passwords within a one-week sprint. Every change is listed in `tests/fixtures/real/aged/README.md`.
4. **One person can reach the lab.** The KDC runs on Ben's home Proxmox server. The tests never need it because the captured fixtures are in the repository; a VirtualBox VM can be built from `docs/lab-setup.md` if needed.
5. **No separate client VM or admin workstation yet.** TC-ENV-01 ran `kinit` on the KDC itself.
6. **Not yet chosen:** the KC-11 default threshold (placeholder 100 requests in 15 minutes) and the OpenAI model.
7. **No automated checks on pull requests.** Tests run when a member runs them; adding a GitHub Actions run is proposed for Sprint 3.
8. **Unpinned packages.** `requirements.txt` uses minimum versions, so TC-PORT-01 may find version differences.

7. Status Summary

Category	Cases	Pass	Fail	Blocked	Not Yet Tested
Environment and fixtures	3	2	0	0	1
Collection and parsing	9	4	0	0	5
Audit rules	15	11	0	0	4
Risk scoring and storage	5	4	0	0	1
AI Gateway	8	0	0	0	8
Reports	4	0	0	0	4
User interface	6	3	0	0	3
Security and non-functional	7	0	0	0	7
Metric evaluation	2	0	0	0	2
Total	**59**	**24**	**0**	**0**	**35**

Sprint 1 ended with 0 of 57 cases run. Sprint 2 ends with 24 of 59 passed, covering every case in Sprint 2 scope. The 35 Not Yet Tested cases are the Sprint 3 work: the log rules, live collection and snapshot export, AI, reports, the remaining screens, and the security, performance and metric tests.

Evidence index. All in `docs/evidence/` on the main branch: `pytest-sprint2.txt` (68 passed), `TC-ENV-01.png`, `TC-DB-01.png`, `TC-KC-13.png`, `TC-UI-02-clean-folder.png`, `TC-UI-02-sample-table.png`, `TC-UI-05.png`, `TC-UI-06-invalid-json.png` and `pytest-53-passed.png` (the run after PR #13).